

Table 4: Comparison with methods on PHOENIX-2014 (the lower the better).

Method	Cue				Dev		Test	
	full	hand	face	pose	del/ins	WER	del/ins	WER
CMLLR (?)		✓	✓	✓	21.8/3.9	55.0	20.3/4.5	53.0
1-Mio-Hands (?)		✓	✓	✓	16.3/4.6	47.1	15.2/4.6	45.1
CNN-Hybrid (?)		✓			12.6/5.1	38.3	11.1/5.7	38.8
SubUNets (?)	✓	✓			14.6/4.0	40.8	14.3/4.0	40.7
Staged-Opt (?)		✓			13.7/7.3	39.4	12.2/7.5	38.7
Re-sign (?)	✓				-	27.1	-	26.8
LS-HAN (?)	✓	✓			-	-	-	38.3
Dilated (?)	✓				8.3/4.8	38.0	7.6/4.8	37.3
Hybrid CNN-HMM (?)		✓			-	31.6	-	32.5
IAN (?)	✓				12.9/2.6	37.1	13.0/2.5	36.7
DenseTCN (?)	✓				10.7/5.1	35.9	10.5/5.5	36.5
CNN-LSTM-HMM (?)	✓	✓			-	26.0	-	26.0
DNF (?) (RGB)	✓				7.8/3.5	23.8	7.8/3.4	24.4
DNF (?) (RGB+Flow)	✓				7.3/3.3	23.1	6.7/3.3	22.9
SMC (ours)	✓	✓	✓	✓	7.6/3.8	22.7	7.4/3.5	22.4
STMC (ours)	✓	✓	✓	✓	7.7/3.4	21.1	7.4/2.6	20.7

Table 5: Comparison with methods on CSL.

Method	Split I	Split II
S2VT (?)	25.5	67.0
LS-HAN (?)	17.3	-
HLSTM-attn (?)	10.2	64.1
CTF (?)	11.2	-
CTM (?)	-	61.9
DenseTCN (?)	14.3	44.7
IAN (?)	-	32.7
STMC (ours)	2.1	28.6

Table 6: Comparison with methods on PHOENIX-2014-T. (f: full-frame, m: mouth, h: hand)

Method	Dev	Test
1 stream (f) (?)	24.5	26.5
2 stream (f+m) (?)	24.5	25.4
3 stream (f+m+h) (?)	22.1	24.1
STMC (ours)	19.6	21.0

works of (?; ?), the dataset is split by two strategies in Table 5. **Split I** is a signer independent test: the train and test sets share the same sentences with no overlap of signers. **Split II** is an unseen sentence test: the train and test sets share the same signers and vocabulary with no overlap of same sentences. Between two settings, **Split II** is more challenging for that recognizing unseen combinations of words is difficult in CSLR. In IAN, their alignment algorithm of CTC decoder and LSTM decoder shows notable improvement, compared with previous methods. Benefiting from multi-cue learning, our STMC framework outperforms the best competitor on CSL by 4.1% on WER.

Evaluation on PHOENIX-2014-T. In Table 6, we provide a result of our method on PHOENIX-2014-T. As a newly proposed dataset (?) for sign language translation, PHOENIX-2014-T provides an extended database with sign gloss annotation and spoken German annotation. CNN-LSTM-HMM utilizes spoken German annotation to infer

the weak mouth shape labels for each video. It provides results of multi-cue sequential parallelism, including full-frame, hand and mouth. Our method surpasses all three combinations of CNN-LSTM-HMM.

5 Conclusion

In this paper, we present a novel multi-cue framework for CSLR, which aims to learn spatial-temporal correlations of visual cues in an end-to-end fashion. In our framework, a spatial multi-cue module is designed with a self-contained pose estimation branch to decompose spatial multi-cue features. Moreover, we propose a temporal multi-cue module composed of the intra-cue and inter-cue paths, which aims to preserve the uniqueness of each cue and explore the synergy of different cues at the same time. A joint optimization strategy is proposed to accomplish multi-cue sequence learning. Extensive experiments on three large-scale CSLR datasets demonstrate the superiority of our STMC framework.

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